2026-10-06 | VS&A Daily

Why matching the SRS is not matching the shock



Learning objective

Explain what a shock response spectrum preserves, identify what it discards, and decide which time-history checks must accompany SRS compliance for a credible test.

The SRS is powerful because it converts a complicated transient into the peak response of a family of damped single-degree-of-freedom oscillators. That is exactly why it is useful for qualification—and exactly why it is incomplete.

Two shocks can produce essentially the same maximax SRS while differing in sign, duration, number of pulses, temporal ordering and cumulative exposure. Linear peak response is matched; the waveform is not reconstructed.

Opening photograph: a technician secures a solar array during NASA Lucy's clampband shock test in March 2021. The test simulated release of the band connecting Lucy to its launch vehicle. Credit: Lockheed Martin; NASA image page, July 20, 2021. Source [4].

Why this matters

Pyroshock is a short-duration, high-frequency structural response generated by devices such as separation nuts, frangible bolts and pin pullers. NASA describes successful shock testing as essential to mission success because propagated shock can damage sensitive hardware [2,3].

The practical trap is treating an SRS overlay as proof that a simulator recreated flight. It proves a narrower statement: for the chosen oscillator model, damping, frequency grid, response metric and processing, the peak responses agree within the stated tolerances.

A bank of peak-hold oscillators

For base acceleration a(t), each oscillator's relative displacement z follows the equation below. The absolute acceleration response is the base motion plus the relative response. An acceleration SRS stores only a selected extreme—often the largest absolute value—over the record.

$$\ddot{z} + 2\zeta\omega_n\dot{z} + \omega_n^2 z = -a(t)$$

$$a_{\text{abs}}(t) = a(t) + \ddot{z}(t) = -2\zeta\omega_n\dot{z}(t) - \omega_n^2 z(t)$$

$$S_{\text{maximax}}(f_n) = \max_t |a_{\text{abs}}(t; f_n, \zeta)|$$

The frequency axis labels the hypothetical oscillators, not the Fourier content of the input. A high SRS value at 3 kHz means that a 3 kHz oscillator reached a large peak under the transient; it does not mean the input contained a stationary 3 kHz tone of that amplitude.

What is discarded

Peak-holding removes when the response occurred and how many comparable peaks followed. A maximax curve also removes response sign. Phase among frequency components, temporal ordering and pulse count are not recoverable. The mapping from time history to SRS is many-to-one, so no unique inverse waveform exists.

Damping and definition matter. Q = 10 corresponds to ζ = 0.05 for the lightly damped oscillator convention Q = 1/(2ζ). A residual SRS, primary SRS, positive/negative SRS and maximax SRS answer different questions. Frequency spacing, sample rate, filtering and record length can also alter the reported curve.

Primary response is the extreme reached while the input is active; residual response is the extreme after the input has ended. Their separation depends on how the active duration is defined. Positive and negative spectra retain directional extrema, while maximax keeps the larger magnitude and discards its sign. For symmetric linear hardware that loss may be acceptable. For unilateral stops, preload-sensitive contacts, latching devices or asymmetric clearances, direction can determine whether a critical state is reached. Record the exact SRS convention in every requirement, prediction and test report; the label 'SRS' alone is underspecified.

NASA-STD-7003A, the publicly accessible 2011 edition, specifies a maximax SRS with Q = 10 for simulated externally induced pyroshock, but it also requires a time history with oscillatory characteristics and duration similar to the predicted flight event, no longer than 20 ms in that context [2]. The point is explicit: SRS compliance does not replace waveform scrutiny. This is NASA-program criteria, not a universal requirement for every program.

One SRS can hide two shocks

The following calculation is illustrative, not measured data. Shock A is a 12 ms decaying multi-frequency pulse. Shock B repeats the identical pulse 120 ms later. Both are sampled at 100 kHz. A bank of absolute-acceleration oscillators from 100 Hz to 5 kHz uses Q = 10.

$$Q = 10 \quad \Longrightarrow \quad \zeta = \frac{1}{2Q} = 0.05$$

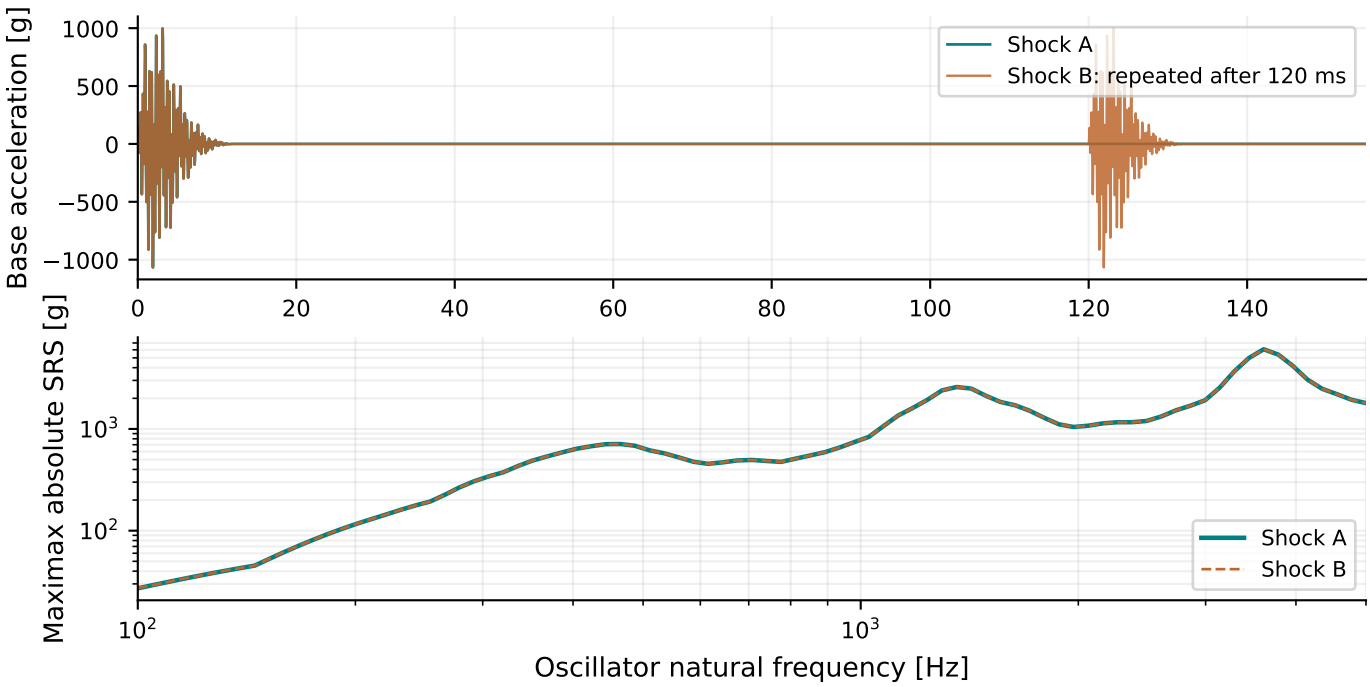


Figure 1. Original numerical example. Shock B contains two separated copies of Shock A. The two maximax SRS curves differ by no more than 0.08 dB on this grid, while the second record is much longer.

The result is not paradoxical. Once the first response has decayed sufficiently, the second identical pulse produces approximately the same peak rather than twice the peak. The maximax operation remembers only the larger excursion. If Shock B were simply -Shock A, their maximax SRS curves would be mathematically identical even though every input polarity is reversed.

A non-peak metric exposes the difference. For this constructed pair, the acceleration-squared integral is:

$$J_a = \int a^2(t)\,dt, \qquad J_{a,B} = 2J_{a,A}$$

Numerically, $J_{a,A}$ = 979 g²·s and $J_{a,B}$ = 1,957 g²·s. This integral is only an exposure descriptor; it is not a universal damage model. It shows that equivalent peak linear response does not imply equivalent accumulated cycling, energy transmission or nonlinear response.

Pair the SRS with observables that retain time

A defensible review starts by stating the failure mechanism. Brittle fracture, relay chatter, connector fretting, PCB strain, fastener slip and nonlinear contact do not necessarily rank waveforms the same way. The SRS oscillator is a screening surrogate, not the hardware itself.

Question	Evidence to retain
Was the source represented?	Device, preload, release sequence, structural path and source distance/field.
Was the waveform credible?	Raw time history, duration, pulse count, sign, crest factor and ring-down.
Was the measurement valid?	Sensor range/resonance, mounting, analog anti-alias filtering, sample rate and clipping checks.
Was the hardware protected?	SRS at critical modes plus local strain, interface loads, continuity and functional monitoring.

Table 1. Original review aid. Applicable evidence depends on hardware, source field and program criteria.

Near-, mid- and far-field behavior

NASA's NESC pyroshock modeling work distinguishes a near field dominated by direct wave propagation, a mid field containing wave propagation and structural resonances, and a far field dominated by lower-frequency bending waves [5]. This is a warning against transferring an SRS through arbitrary scalar attenuation alone. Joints, geometry, materials and distance reshape the transient.

Measurement artifacts can be especially persuasive because they often inflate the high-frequency SRS. Inspect unfiltered data for overload and zero shift, then verify the complete analog and digital signal chain. NASA-STD-7003A requires analog anti-alias filtering before digitization and preservation of the time history for comparison [2]. Those requirements belong to that standard's scope; the underlying signal-integrity logic is broadly useful.

Use engineering units and bandwidth consistently when comparing locations. A sensor mounted on a small bracket can report a local resonance that is not representative of the equipment interface; a heavily filtered channel can suppress a real near-field peak. Triangulate with multiple channels, mounting orientations and structural knowledge. When a waveform looks nonphysical, investigate the measurement before relaxing or raising the requirement. Both a false high and a false low can make the qualification evidence unsafe.

Finally, examine tolerances where the hardware is sensitive. A broad compliance band can permit undertest at a vulnerable mode and overtest elsewhere. NASA's 2024 discussion of updates to NASA-STD-7003B highlights tighter attention at hardware resonances and cautions that some shaker simulations cannot reproduce the full shock bandwidth or representative signatures [3].

Use SRS as a compression, not a reconstruction

The SRS is exceptionally effective for comparing short transients against families of linear resonators. It supports specification, statistical enveloping and test control without requiring a detailed model of every component. Its limitation is not a flaw; it is the consequence of compressing a time history into one peak per oscillator.

Reliability improves when the acceptance argument preserves both views. The SRS protects peak modal response over frequency. The time history protects source realism, pulse duration, polarity, sequencing, repeated events and instrumentation integrity. Local response measurements protect failure modes that the generic oscillator bank cannot represent.

Three takeaways

1. Matching SRS is a bounded claim. It demonstrates similar peak response for a specified oscillator bank, damping, metric and processing—not equivalent waveforms.

2. Keep the time domain. Review duration, pulse count, polarity, ring-down, clipping and oscillatory character alongside the SRS.

3. Anchor the test to failure physics. Use representative source paths and monitor the local quantities—strain, load, continuity or function—that connect shock exposure to hardware risk.

Reflection

Two test records meet the same maximax SRS tolerance. One contains a single 10 ms pulse; the other contains three separated pulses. Can they be declared equivalent? **Answer:** not from the SRS alone. Check flight-event count and duration, cumulative or nonlinear susceptibility, waveform realism, and the governing program criteria. The extra pulses may add exposure without raising the peak-hold spectrum.

Sources and scope

[1] W. O. Hughes and A. M. McNelis, Pyroshock Environments Characterized for Spacecraft Missions, NASA Lewis Research Center, April 1999. Definition, mission significance and EOS separation-test database. NASA NTRS.

[2] NASA-STD-7003A, Pyroshock Test Criteria, December 20, 2011, especially §§1.6 and 4.4-4.8. Publicly accessible edition; requirements apply within its defined NASA scope and are not asserted here as universal. NASA PDF.

[3] NASA, Mentoring the Next Generation of Engineers and Improving Shock Testing Standards, March 7, 2024. Discussion of NASA-STD-7003B application and proposed future clarifications. NASA article.

[4] NASA, NASA's Lucy Spacecraft Gets Shocked, July 20, 2021. Photograph credit: Lockheed Martin. Image and context. [5] S. J. Gentz et al., Empirical Model Development for Predicting Shock Response on Composite Materials Subjected to Pyroshock Loading, NASA/TM-2015-218781 Vol. I, July 2015. NASA NTRS.

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